Elastic Monetary Architecture for BRICS

Empirical evidence, governance via λ, non-linear reconvergence, and clearing-anchored endogenous supply

Armando José Freire de Melo

Consolidated document — 2026-03-03

This PDF consolidates, in a single file, the academic documentation and replication appendices based on the author's scripts and manuscripts. The opening sections (Reproduction Guide + Annex A) were assembled from the author's outline; afterwards, the document incorporates the original manuscripts as supplementary parts, preserving figures and tables.

Abstract

This document consolidates a monetary-design framework for an emerging bloc, focusing on BRICS: (i) empirical evidence that aggregating real output reduces growth volatility relative to individual members; (ii) a non-linear elastic reconvergence mechanism that pulls the price toward a macroeconomic fundamental, with an explicit local stability condition $0 < \alpha k < 2$; (iii) a governance dial λ that parameterizes concentration of GDP^{λ} weights, measured by HHI and the dominant member share; (iv) a stepwise institutional layer (reported fundamental updated in windows) and windowed supply-policy regimes (mirror burn β and an amortized regime with structural emission); and (v) an operational alternative for endogenous supply anchored in clearing activity (mini-GDP), with EMA smoothing and adjustment caps.

Organized technical contributions

• Stability mechanism with an engineering restriction $0 < \alpha k < 2$ and empirical tests (φ estimation, basin of attraction, and global convergence).

• Governance measurable via λ: concentration (HHI) and dominance (China share) connected to resilience under a shock to the core.

• Institutional staircase: separation between continuous real-economy evolution and the fundamental observed in reporting windows.

• Windowed policy regimes (mint/burn): explicit trade-off between countercyclical aggressiveness and risk (volatility, drawdown).

• Clearing-based alternative anchor: mini-GDP derived from intra-bloc volumes, with EMA smoothing and an annual CAP to avoid runaway.

Reading note: after the Reproduction Guide and Annex A, this PDF includes the manuscripts and notes in their original format, preserving internal figure/table numbering and pagination for each part.

Reproduction guide and script mapping

Below is the minimal map to reproduce the figures and experiments that support the manuscript. Names correspond to the Python files provided by the author and the PDFs included in this document.

Block / Section	What it demonstrates	Main file(s)
Empirical evidence (2000–2024)	Aggregate growth volatility vs members; λ sensitivity; Var/Cov decomposition; correlations.	Estabilidade_Macroeconomica_BRICS_2000_2024 (1).pdf
Elastic mechanism (full paper)	Non-linear reconvergence; local stability $0 < \alpha k < 2$; Figures 1–10; price vs fundamental.	paper_structural_lambda_full.py
Scenario A — macro shock	Shock to a subset of countries; metrics (vol, MDD, recovery, re-anchoring).	cenario_A_choque_macro_lambda.py
Scenario B — 2008-style crisis	Shock + temporary volatility spike; comparison with GBM benchmark.	cenario_B_crise_2008_lambda.py
Scenario C — core shock	Sensitivity of volatility and drawdown to λ under a shock to the dominant member.	cenario_C_lambda_sweep.py
Stepwise supply: mirror burn	Burn proportional to reported contraction; countercyclical defense controlled by β.	sistema1.py
Stepwise supply: 2% amortized	Structural emission + partial burn; separates trend from countercyclical defense.	sistema2.py
Clearing: mini-GDP & endogenous supply	$g_t = \log(V_t/V_{t-1})$, EMA, annual CAP; monthly supply adjustment.	Clearing.py; Paper_2_Clearing_Endogenous_Supply.pdf
Empirical stability tests	φ estimation, basin of attraction, and global convergence; version with λ embedded.	teste_estrutura_global.py; teste_lambda_estrutural.py

Note: supplementary parts preserve the original figure/table numbering and internal pagination for fidelity.

Code and materials availability

Source code, replication scripts, and supplementary materials associated with this document are publicly available in the author's repository (GitHub):

https://github.com/armanfm/brics-elastic-monetary-model-

Suggested citation (software/replication):

Melo, A. J. F. (2026). BRICS Elastic Monetary Model — simulation code and replication materials. GitHub repository.

Annex A — Sensitivity to λ in BRICS vs FIAT (USD/EUR)

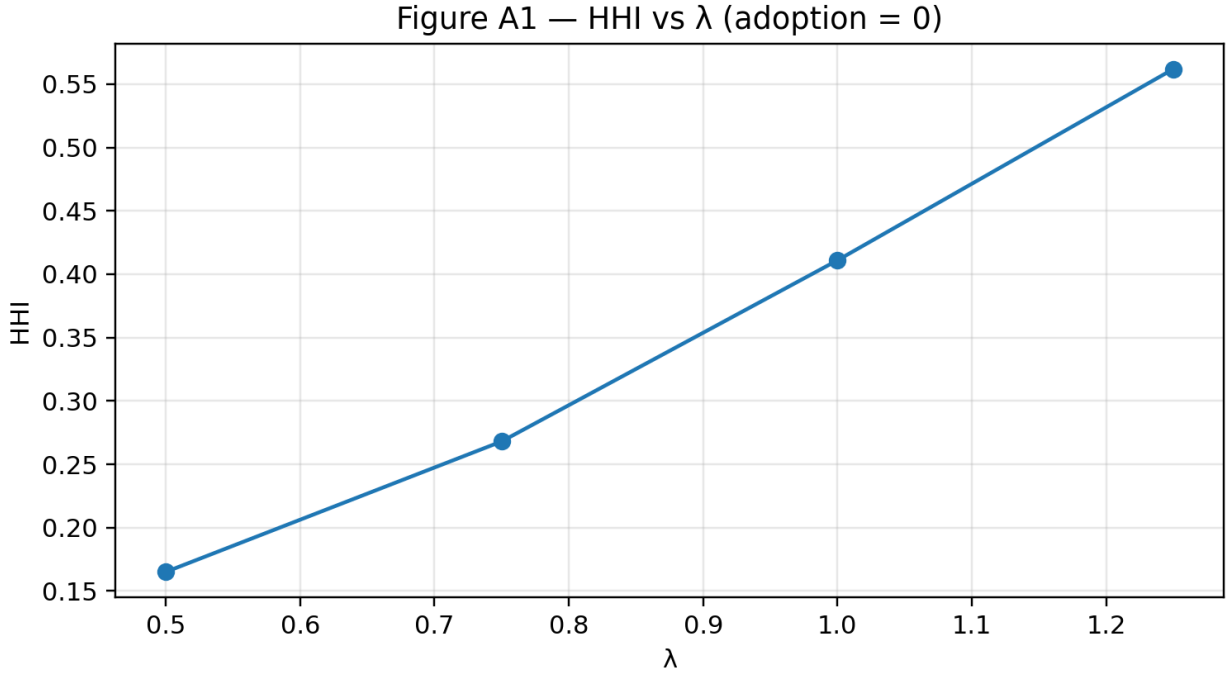
This annex runs a grid of BRICS vs FIAT simulations with logistic adoption (levels 0, 0.5, 1.0) to quantify how governance via λ (concentration and dominance) relates to risk/return of the elastic engine and to end-horizon outperformance probabilities.

Configuration: horizon = 1260 days (~5 years), 150 simulations per combination; α=0.5, k=2.0; vol_espec_base=0.04, intensidade_espec_base=0.6, vol_ativos_base=0.10; logistic adoption with g=0.02 and t0=630; FIAT with explicit hypotheses: USD (μ=2.5%/yr, σ=6%/yr), EUR (μ=2.0%/yr, σ=7%/yr).

λ	Adoption	HHI	China (%)	Ret_BRICS (%)	Vol_BRICS (%)	Sharpe	MDD (%)	P(B>USD) (%)	P(B>EUR) (%
0.50	0.0	0.165	31.9	2.84	10.04	0.28	-19.7	53.3	56.0
0.50	0.5	0.165	31.9	2.39	9.28	0.26	-18.9	52.0	56.7
0.50	1.0	0.165	31.9	2.87	8.63	0.33	-16.8	55.3	60.0
0.75	0.0	0.268	47.1	2.43	10.01	0.24	-19.9	50.0	54.0
0.75	0.5	0.268	47.1	2.56	9.29	0.27	-18.2	57.3	54.0
0.75	1.0	0.268	47.1	2.23	8.61	0.26	-17.8	46.0	55.3
1.00	0.0	0.411	61.7	2.79	10.05	0.28	-19.5	56.7	58.7
1.00	0.5	0.411	61.7	2.96	9.30	0.32	-17.8	55.3	61.3
1.00	1.0	0.411	61.7	2.86	8.60	0.33	-17.0	56.7	56.0
1.25	0.0	0.562	73.8	3.02	10.05	0.30	-19.3	59.3	57.3
1.25	0.5	0.562	73.8	2.80	9.28	0.30	-17.9	56.0	58.7
1.25	1.0	0.562	73.8	2.76	8.64	0.32	-17.0	62.7	55.3

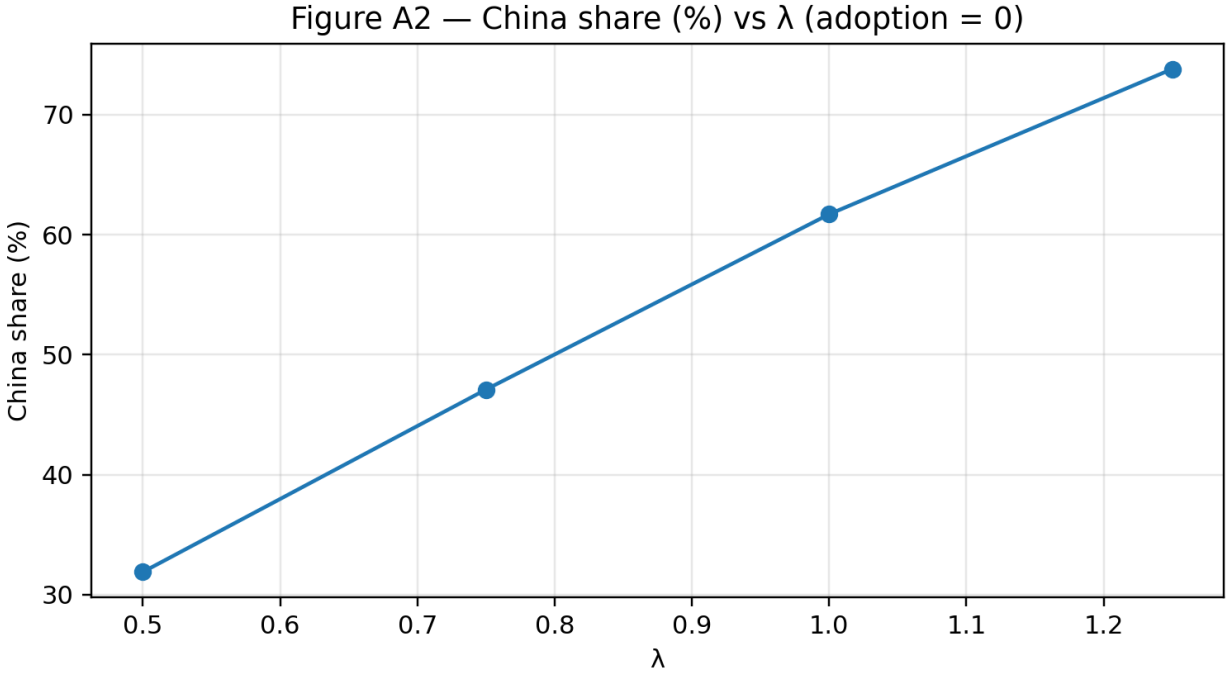
Table A1 — Grid means (elastic BRICS). Return and volatility are annualized; MDD is the mean maximum drawdown (in %).

Figure A1 — HHI vs λ



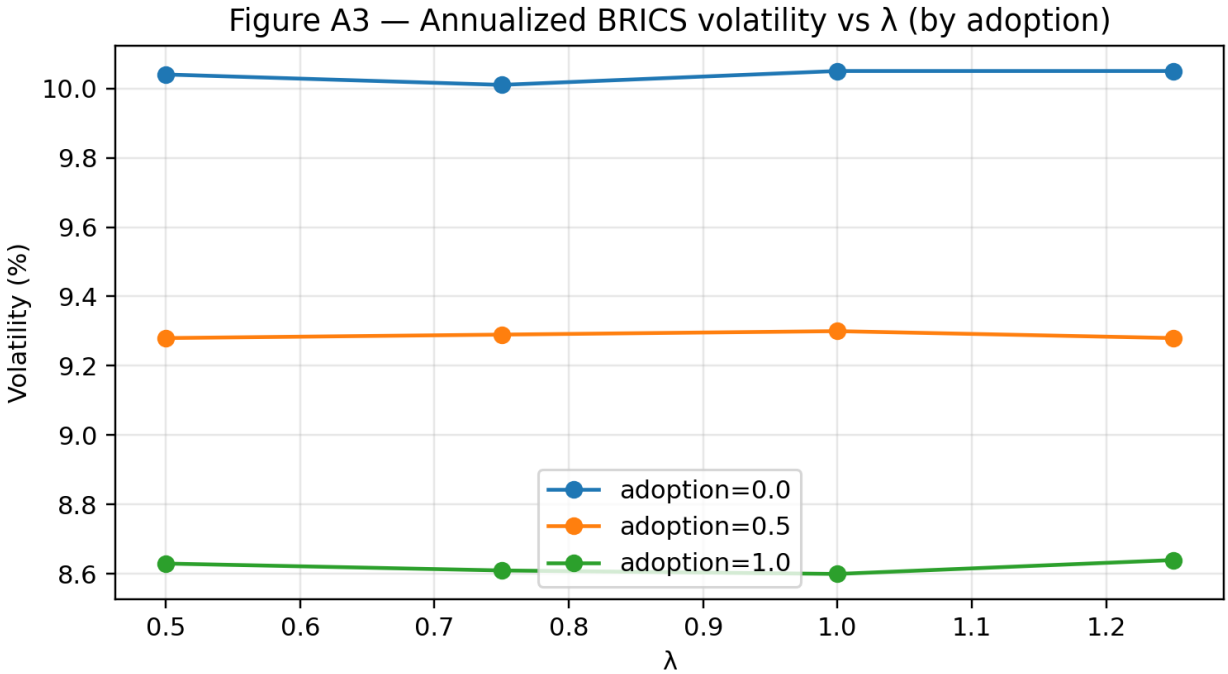
Reading: As λ increases, structural concentration rises (HHI rises) and the dominant member share increases, shifting the governance-resilience trade-off.

Figure A2 — China share (%) vs λ



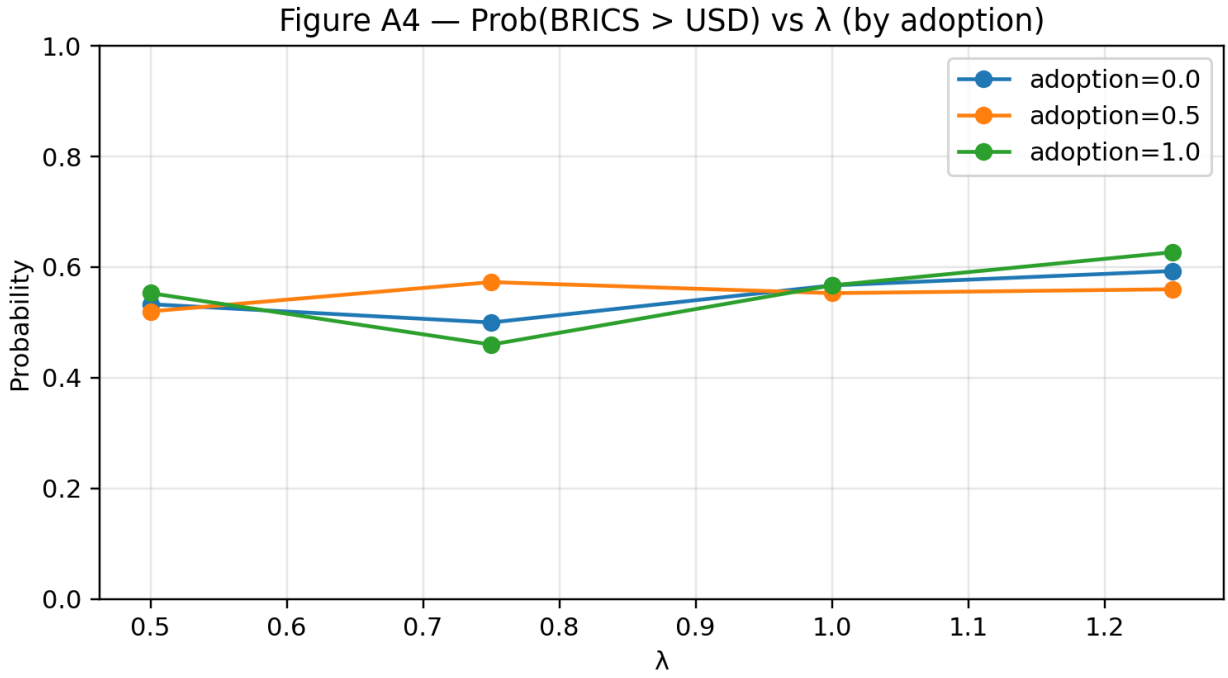
Reading: The dominant member share is an increasing function of λ, enabling transparent parameterization of dominance.

Figure A3 — BRICS volatility vs λ (by adoption)



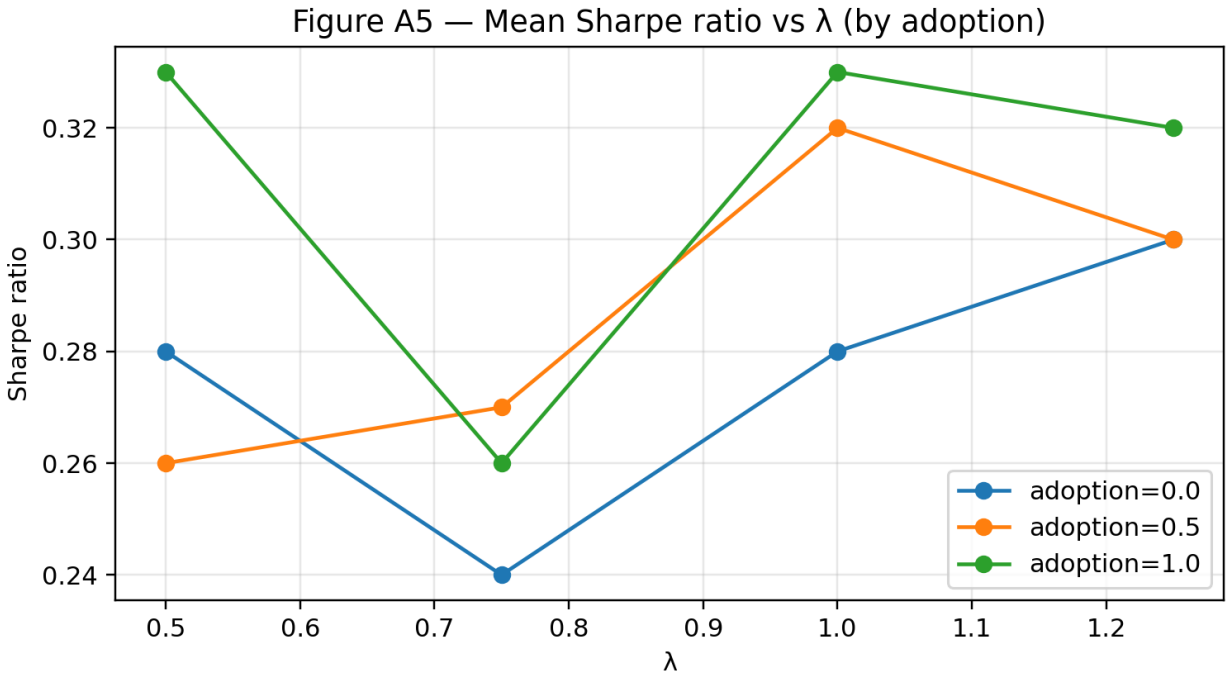
Reading: Adoption reduces volatility by damping both the real and speculative components; λ changes concentration structure.

Figure A4 — Prob(BRICS > USD) vs λ (by adoption)



Reading: This metric summarizes the probability of the BRICS engine outperforming the FIAT benchmark under explicit μ/σ assumptions.

Figure A5 — Sharpe vs λ (by adoption)



Reading: Sharpe consolidates risk/return; λ and adoption illustrate how governance and penetration affect bloc efficiency.

BRICS Elastic Monetary Architecture with Clearing-Based Activity Anchors and a Governance Reserve

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Abstract. We integrate (i) empirical evidence that aggregation smooths real-output volatility within the BRICS bloc and (ii) a rule-based monetary design for a basket unit featuring non-linear elastic reconvergence toward a macro fundamental. Governance is parameterized by a concentration dial λ (GDP^{λ} weights), with concentration measured by HHI and the dominant member share. We extend the baseline with stepwise reporting, stress tests, supply regimes (mirror burn β and an amortized 2% annual structural emission), and an implementable institutional layer where a clearing authority holds genesis supply S0 and executes mint/burn in a governance reserve (no automatic redistribution). We also show how clearing volumes define a monthly activity proxy (mini-GDP) for supply updates with smoothing and caps.

Keywords: BRICS; clearing unions; basket currencies; monetary design; stability; governance.

JEL: E42; E58; F33; F42.

1. Introduction

The practical problem addressed by this manuscript is institutional: how can a heterogeneous economic bloc build a settlement layer that is more stable than its members taken individually, while avoiding both full monetary union and excessive structural dominance by the largest economy. The approach is engineering-driven: we specify an explicit macro fundamental, a market-price component with noise, and a deterministic reconvergence rule with a closed-form local stability restriction. We then connect governance and resilience through a single dial (λ) and investigate stress behavior under discrete reporting and crisis-style volatility spikes. Finally, we translate the mechanism into an implementable clearing architecture with an authority-managed governance reserve.

2. Empirical evidence: stability of aggregated output

We summarize empirical evidence (2001-2024 growth rates derived from World Bank WDI real GDP in constant 2015 USD) showing that aggregated BRICS output growth is smoother than individual members. The core observation is volatility smoothing under aggregation and the role of weighting rules. We also provide an approximate 'BRICS without China' robustness diagnostic and a jackknife-style exclusion exercise based on second moments.

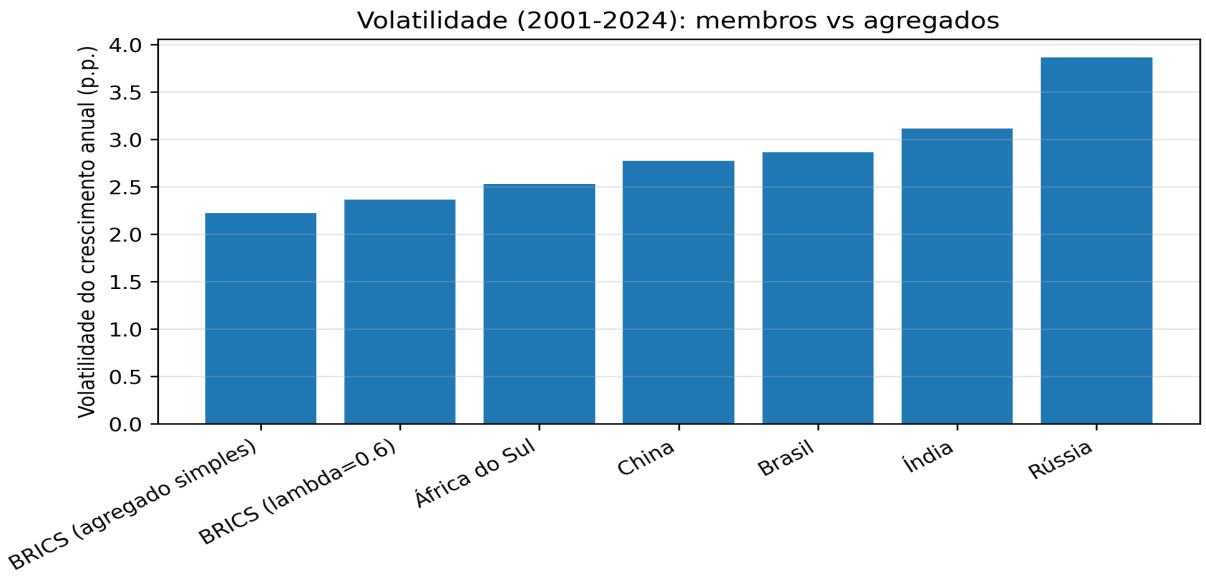

Figure 1. Empirical volatility of annual real GDP growth: members vs. aggregates.

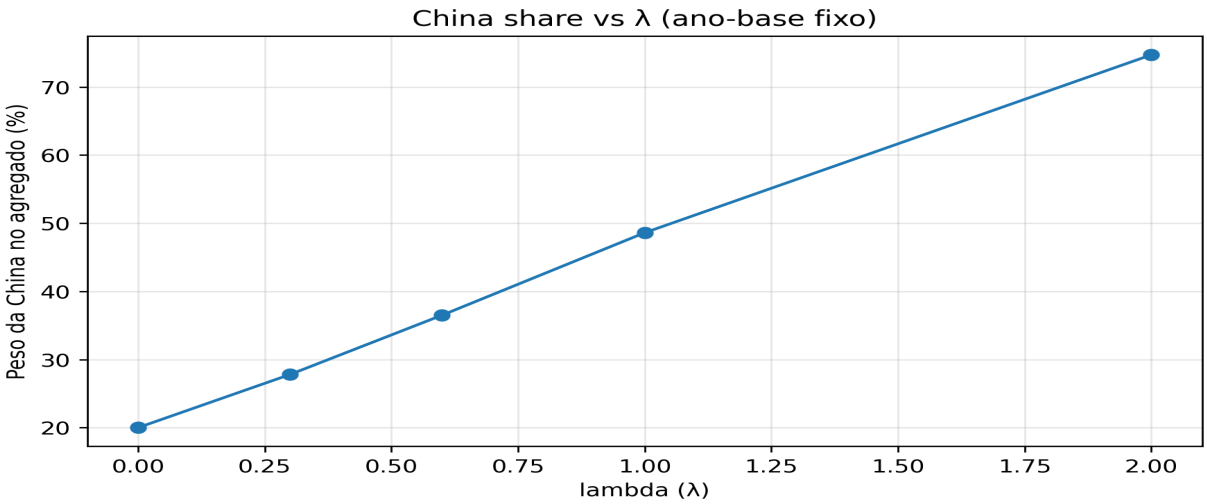

Figure 2. China share as a function of λ (empirical).

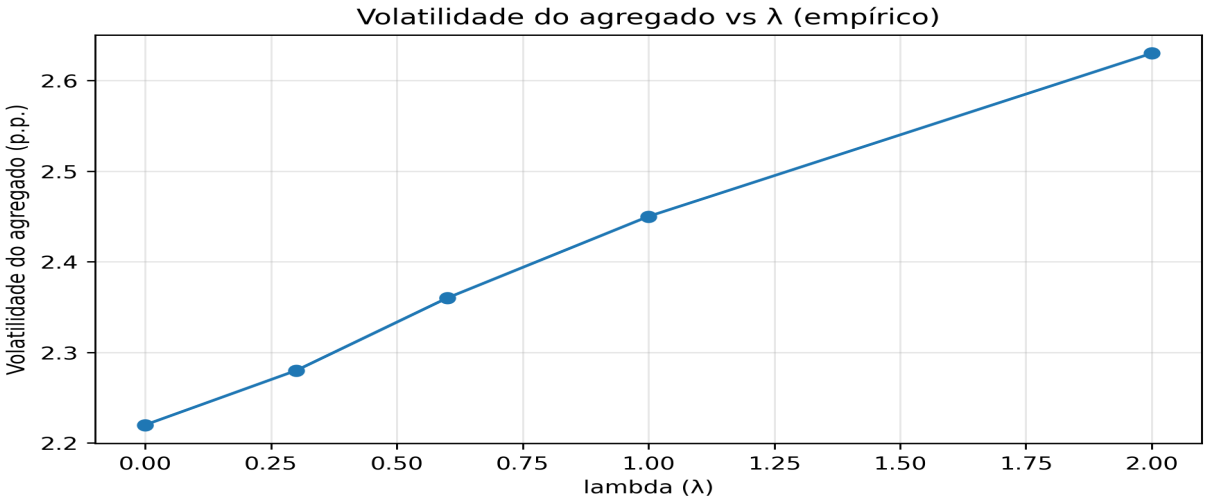

Figure 3. Aggregate volatility as a function of λ (empirical).

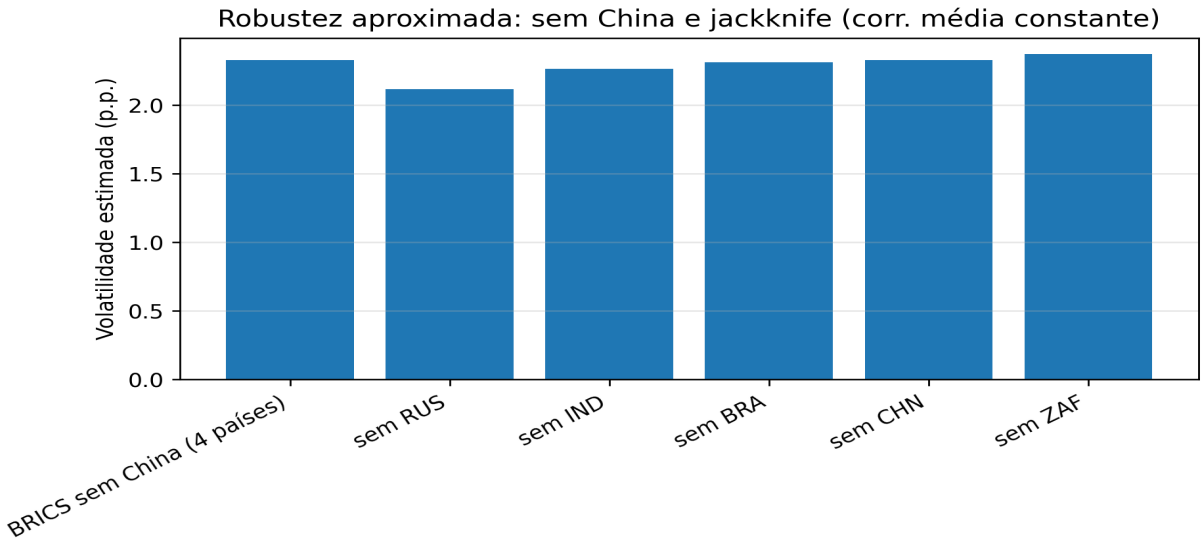

Figure 4. Robustness approximation: aggregate without China and jackknife (constant average correlation hypothesis).

3. Model: elastic reconvergence toward a macro fundamental

We define a macro fundamental as the ratio between a reported effective capital index and supply. Reported capital is updated in discrete windows to reflect real-world reporting frequency. Market price is subject to multiplicative speculation, and a non-linear reconvergence term corrects deviations.

$$F_t = \frac{\widehat{C}_t}{S_t}$$

$$P_t = (1-\alpha)P_{\text{spec},\,t} + \alpha\, P_{\text{spec},\,t}\left(\frac{F_t}{P_{\text{spec},\,t}}\right)^k$$

Local stability restriction (engineering constraint):

$$0 < \alpha k < 2$$

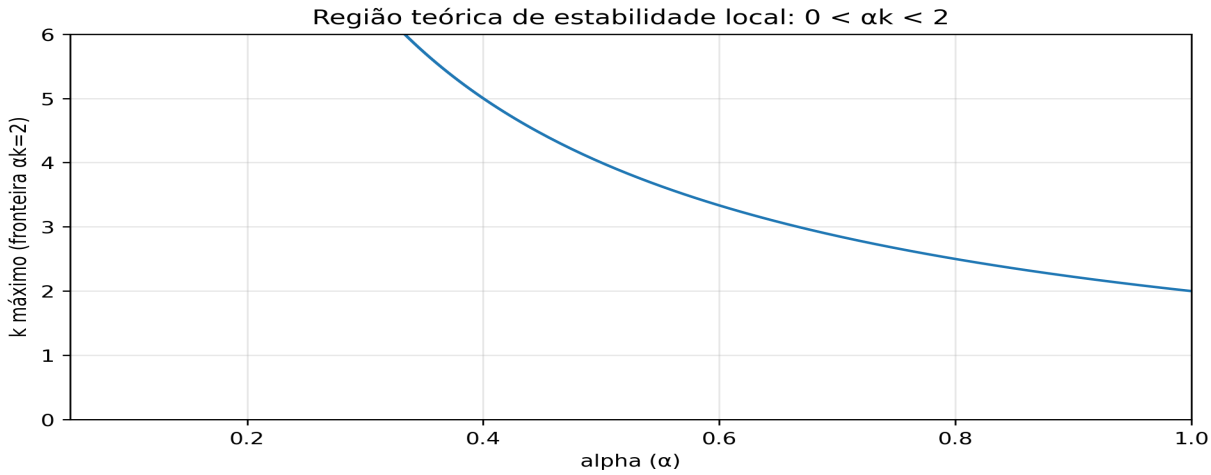

Figure 5. Local stability region: 0 < αk < 2.

3.1 Stepwise reporting ('staircase' fundamental)

To model institutional reporting frequency, we distinguish the latent daily evolution of capital from the reported anchor used by the monetary rule. The reported fundamental is updated on discrete windows (e.g., quarterly) and remains constant between releases. This produces characteristic plateaus with elastic re-anchoring.

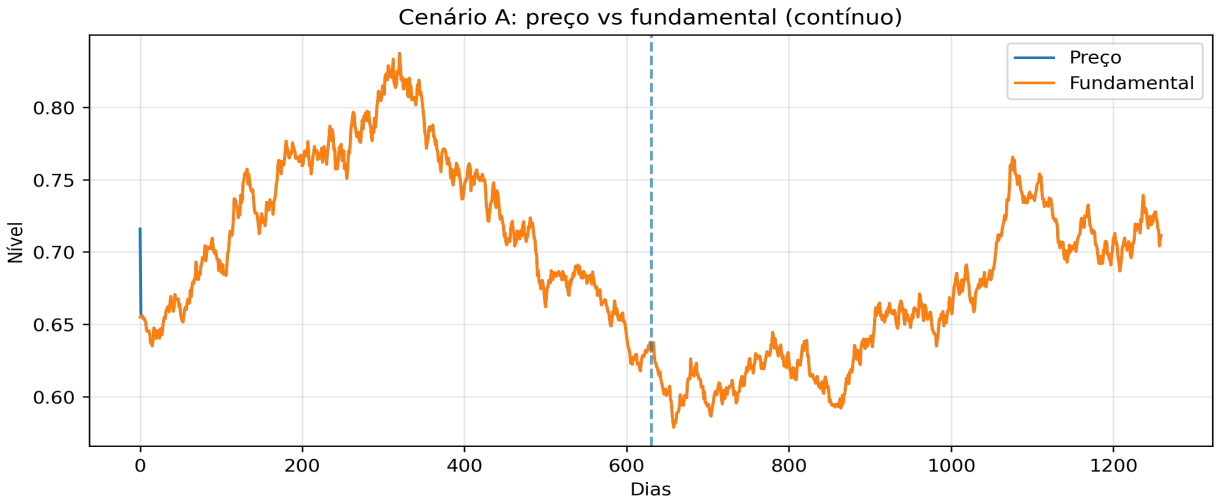

Figure 6. Scenario A: price vs. continuous fundamental (example path).

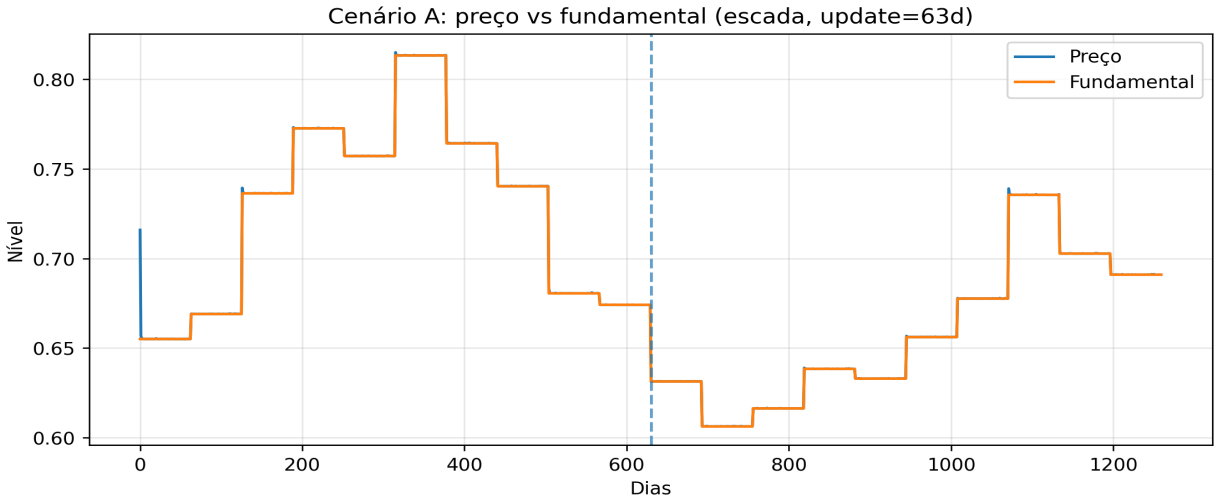

Figure 7. Scenario A: price vs. stepwise reported fundamental (quarterly update).

4. Governance and concentration: the λ parameter

Governance weights are parameterized by a power rule on economic size. The parameter λ controls structural concentration. Lower λ distributes weights more evenly; higher λ concentrates influence in the largest members. We quantify concentration using HHI and report the dominant member share.

$$w_i = \frac{\mathrm{GDP}_i^{\lambda}}{\sum_j \mathrm{GDP}_j^{\lambda}}$$

Table 1. Governance weights under λ = 0.5 (GDP proxy). HHI = 0.165.

Member	Weight (%)
CH	31.94
IN	14.48
RU	11.17
BR	10.91
SAU	7.90
IR	5.83
UAE	5.32
ZA	4.76
EG	4.76
ET	2.92

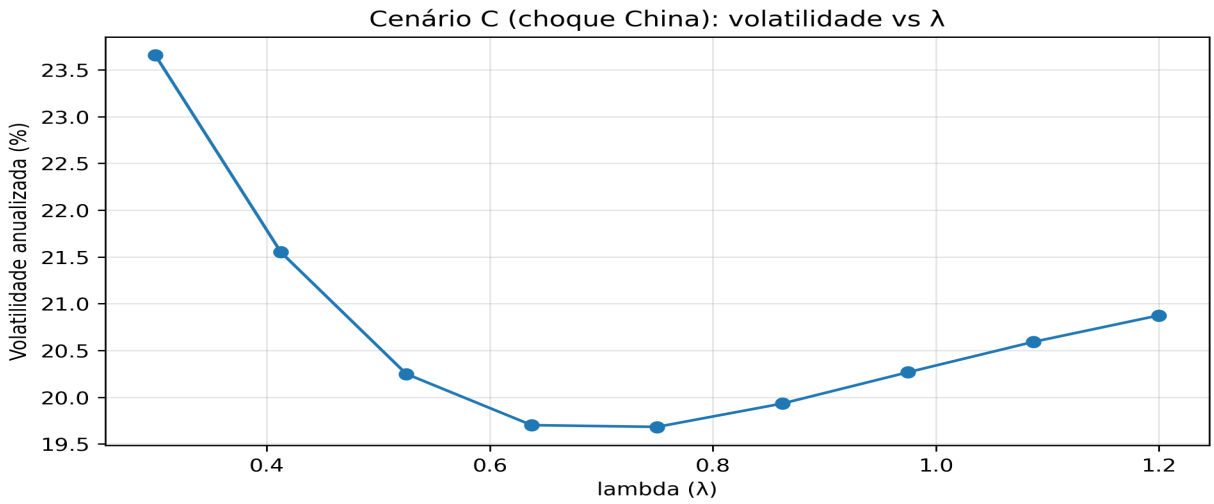

Figure 8. Scenario C (shock to dominant member): volatility vs. λ (Monte Carlo).

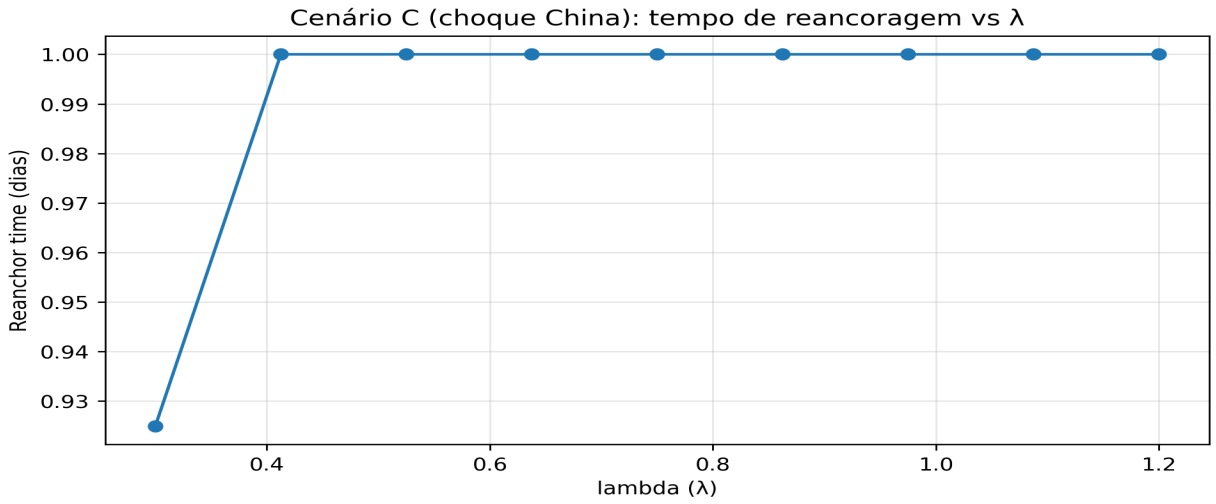

Figure 9. Scenario C: re-anchoring time (<= 5% band) vs. λ (Monte Carlo).

5. Stress tests

We evaluate two stress scenarios: (A) a macro shock applied to a subset of members and (B) a crisis-style scenario with a larger shock and a temporary volatility spike. Benchmarks include a macro GBM process. The figures report typical paths and summary risk metrics.

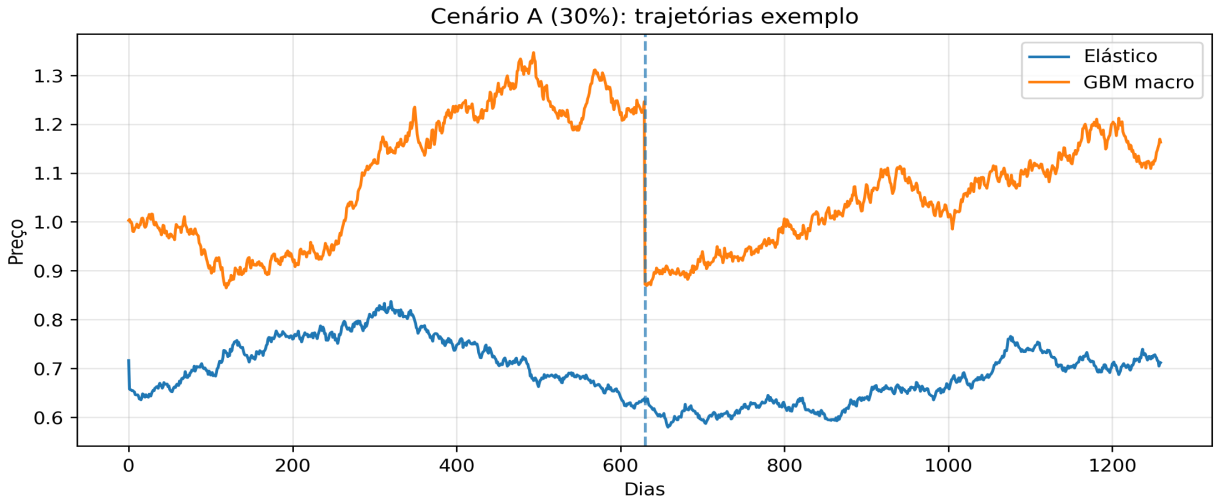

Figure 10. Scenario A: example trajectories (elastic vs. benchmark).

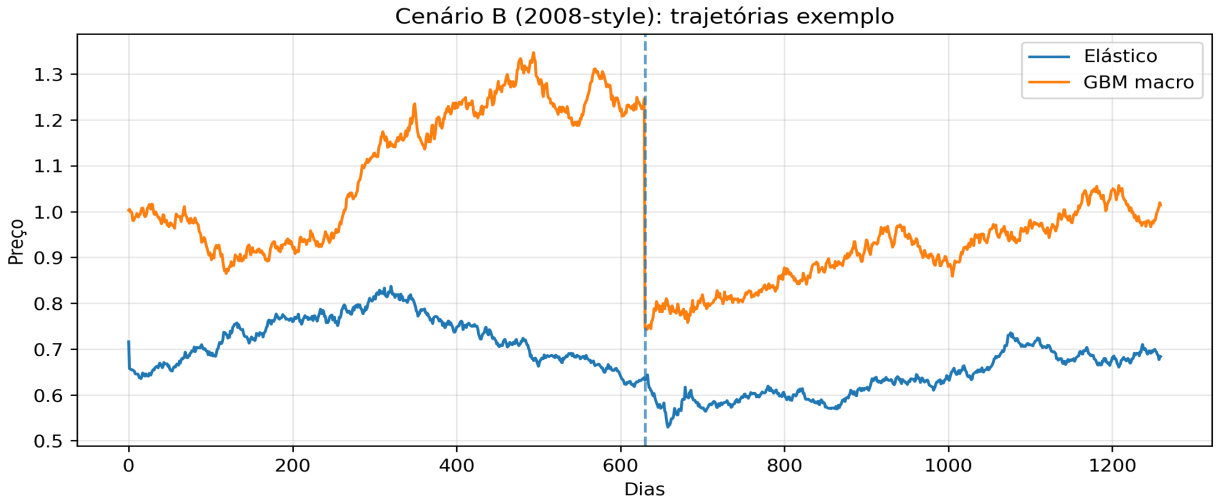

Figure 11. Scenario B (crisis-style): example trajectories.

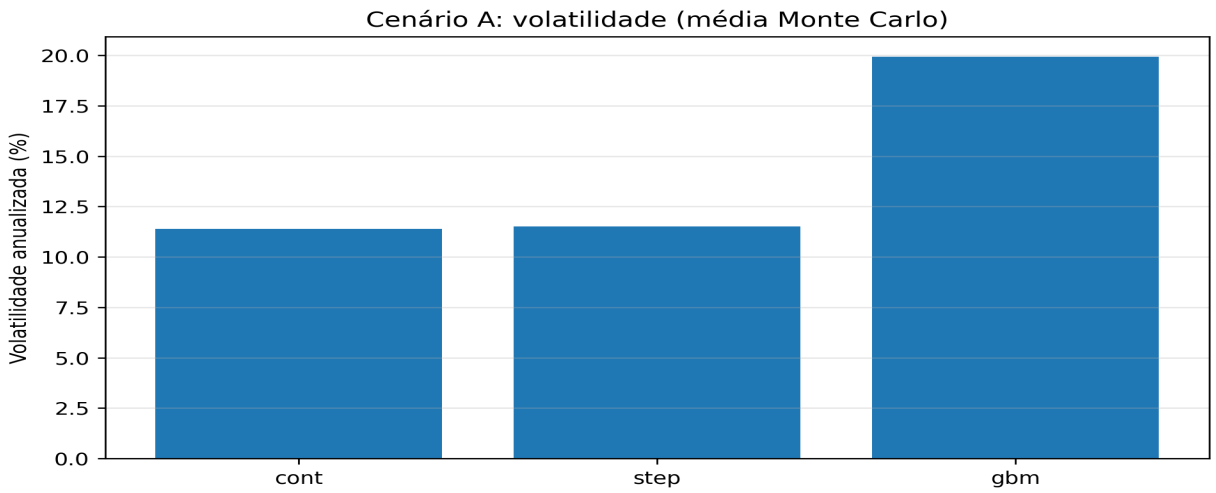

Figure 12. Scenario A: annualized volatility by model (mean Monte Carlo).

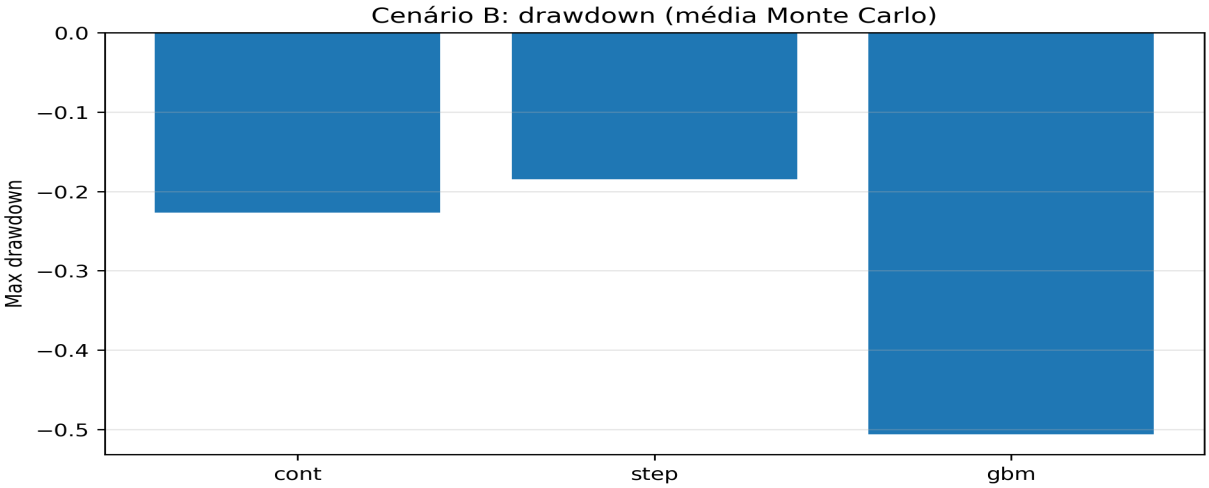

Figure 13. Scenario B: max drawdown by model (mean Monte Carlo).

6. Policy regimes for stepwise supply adjustment (mint/burn)

We extend the stepwise architecture to the supply itself. Under windowed updates, we compare: (i) no burn, (ii) mirror burn proportional to the observed contraction with intensity β in [0,1], and (iii) an amortized rule with a 2% annual structural emission plus partial burn when contraction is detected. This section isolates the trade-off between resilience and policy aggressiveness.

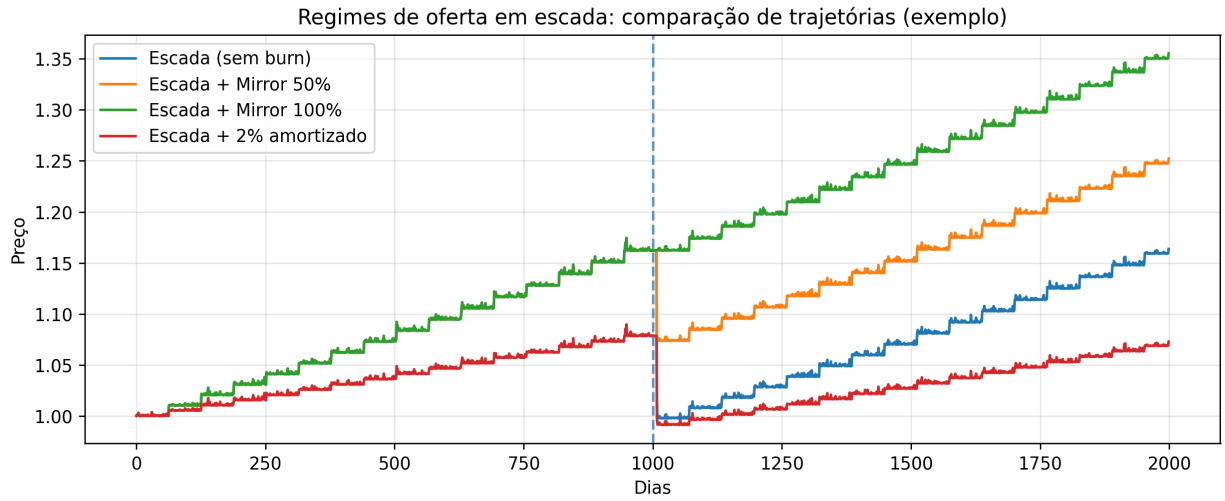

Figure 14. Example trajectories under stepwise supply regimes.

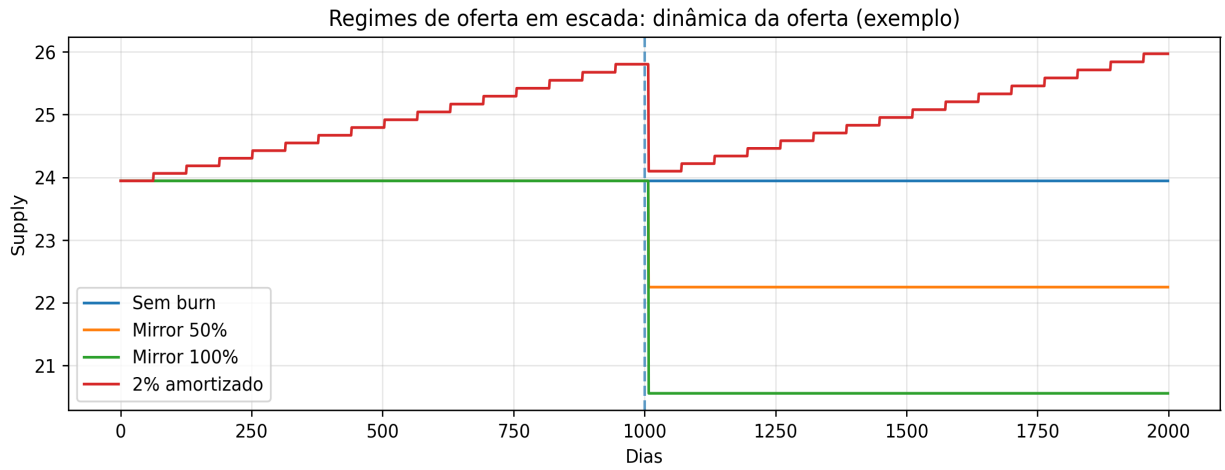

Figure 15. Supply dynamics under stepwise regimes (example).

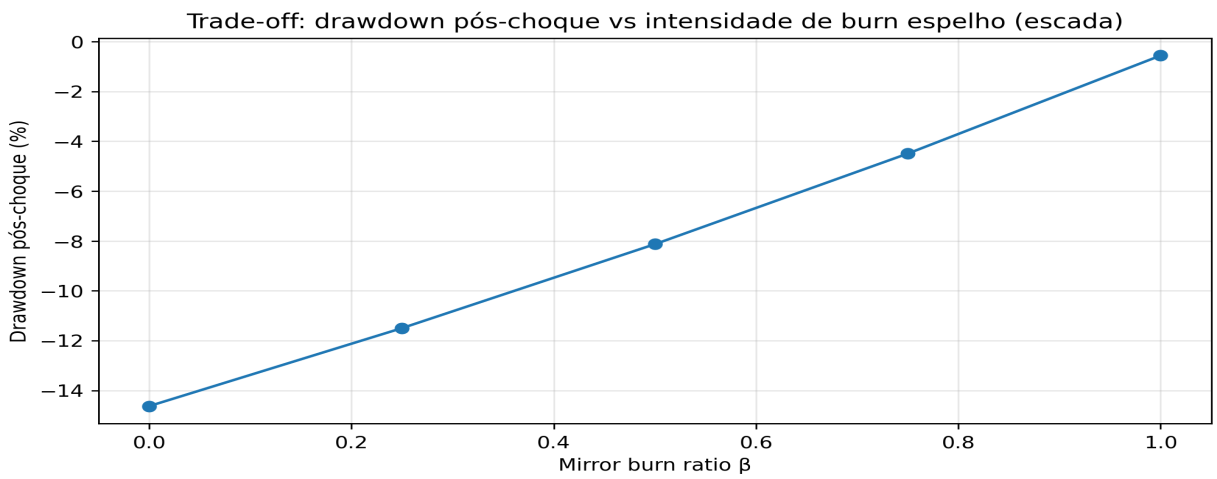

Figure 16. Post-shock drawdown vs. mirror burn intensity β (Monte Carlo).

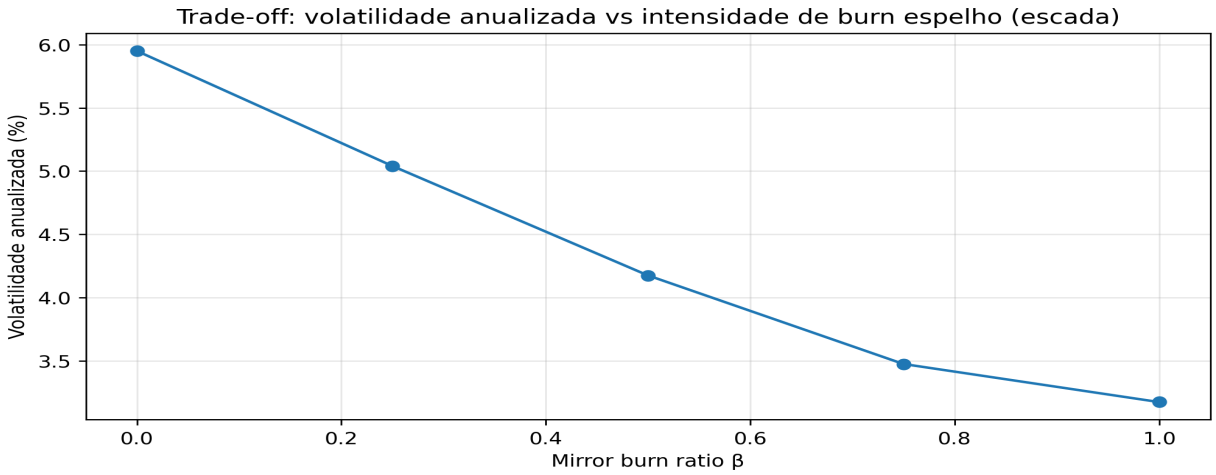

Figure 17. Annualized volatility vs. mirror burn intensity β (Monte Carlo).

Table 2. Summary of stepwise supply regimes (means from Monte Carlo; shock=15%, update=63 days).

Regime	Vol (%)	MDD (%)	DD_shock (%)	Reanchor (d)
Sem burn (β=0)	5.95	-14.64	-14.63	0.0
Mirror β=0.25	5.04	-11.51	-11.50	0.0
Mirror β=0.50	4.18	-8.14	-8.13	0.0
Mirror β=0.75	3.48	-4.51	-4.50	0.0
Mirror β=1.00	3.17	-1.01	-0.56	0.0
2% amortizado (emissão 2% a.a., burn fator=0.5)	3.94	-8.60	-8.59	0.0

7. Clearing-based activity anchors (mini-GDP) with monthly updates

To improve operational feasibility, the model can replace low-frequency macro releases with a high-frequency activity proxy computed from observed intra-bloc clearing volumes. We define a mini-GDP growth signal from clearing volume V_t, smooth it via EMA, and update supply on monthly windows under caps to avoid runaway behavior.

$$g_t^{\text{clear}} = \ln\left(\frac{V_t}{V_{t-1}}\right)$$

$$S_{t+1} = S_t \exp(\beta\, g_t^{\text{EMA}}), \quad |\Delta \ln S| \le \text{CAP}$$

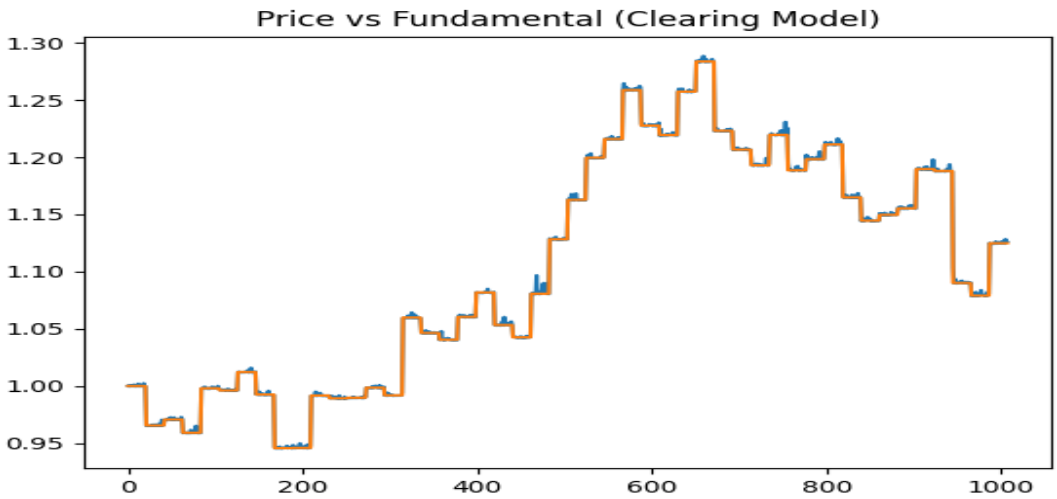

Figure 18. Clearing-based model: price vs. stepwise fundamental under monthly updates (example).

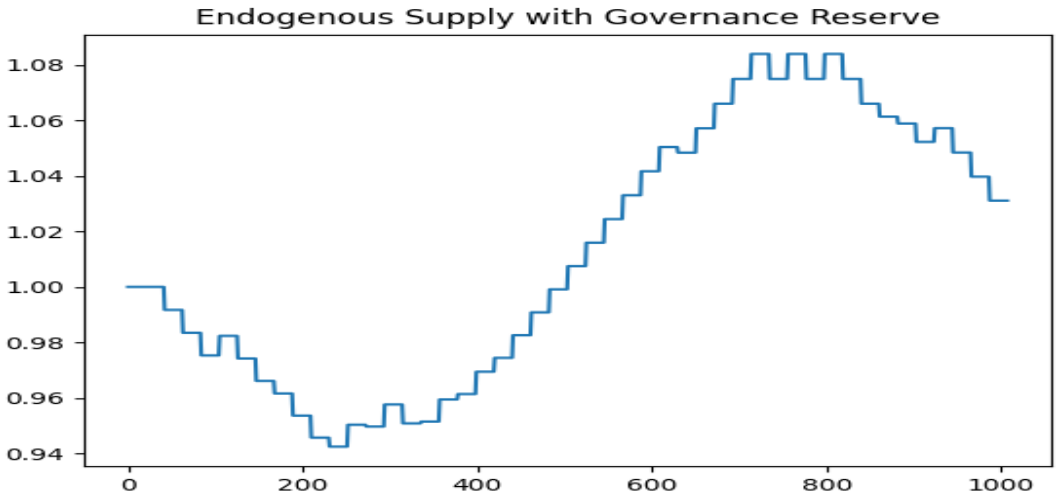

Figure 19. Clearing-based model: endogenous supply path under the mini-GDP rule (example).

8. Institutional implementation: genesis supply and governance reserve

We propose an implementable clearing architecture that avoids political redistribution. A BRICS Clearing Authority (BCA) holds the genesis supply S0 in a reserve account. Members and authorized institutions acquire the unit through demand-based exchange (paying in domestic currency or approved reserve assets), and redeem it symmetrically. Rule-based mint and burn operations, when used, are executed within a governance reserve controlled by the BCA rather than distributed across member states.

Genesis and custody. Total supply S0 is created at inception and held by the BCA reserve account. Member accounts start at zero.

Acquisition/redemption. A member acquires units by transferring domestic currency/reserve assets to the BCA, which transfers units from its reserve. Redemption reverses the operation.

Governance reserve. If the rule increases supply, newly created units accrue to the governance reserve (not to members). If the rule reduces supply, it is burned from that reserve. This avoids implicit cross-country transfers.

9. Long-run drift control

Under structural growth g_real of the reported capital index and structural monetary growth g_monetary (e.g., a fixed emission schedule), the long-run drift of the fundamental satisfies an approximate identity. This provides a simple policy knob for long-run appreciation, neutrality, or inflation regimes.

$$\frac{d}{dt}\ln F_t \approx g_{\text{real}} - g_{\text{monetary}}$$

10. Limitations

The simulations are stylized and intentionally transparent. Capital is proxied as a scaled function of GDP, cross-country correlations are simplified, and market microstructure/liquidity frictions are abstracted. Institutional implementation would require agreed reporting standards for clearing volumes, anti-manipulation controls (e.g., excluding circular transactions), and explicit credit-limit and settlement policies for persistent imbalances.

11. Conclusion

We provide a cohesive design space for a BRICS basket settlement unit: a macro fundamental with elastic reconvergence, a governance dial λ that quantifies concentration, a stepwise reporting layer aligned with real institutional data releases, stress-tested supply policies (mirror burn β and amortized emission), and a clearing-based mini-GDP anchor enabling monthly updates with smoothing and caps. An implementable institutional framework places genesis supply and rule-based mint/burn within a governance reserve held by a clearing authority, avoiding political redistribution while preserving domestic monetary sovereignty.

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A Clearing-Based Endogenous Supply Mechanism for a Multilateral Trade Unit

Abstract

This paper presents an operational multilateral trade unit anchored in clearing-based activity. Supply evolves according to a smoothed growth rate derived from intra-bloc clearing volume (mini-GDP), with bounded adjustment constraints. Governance is parameterized through λ-power aggregation. Simulations confirm structural stability and price-fundamental coherence.

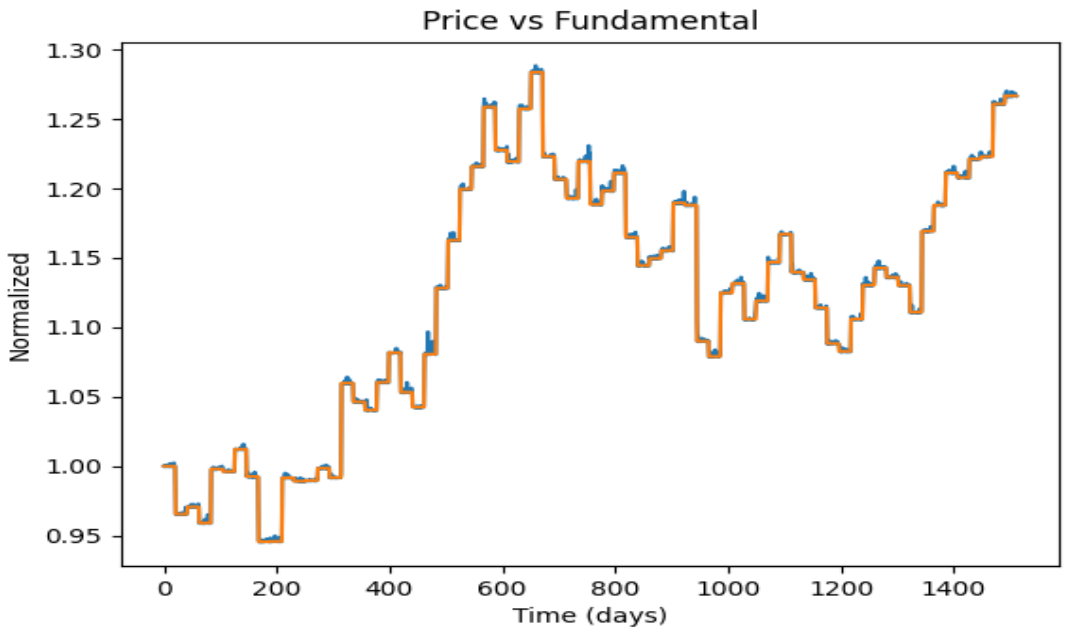
1. Governance Structure ($\lambda = 0.5$)

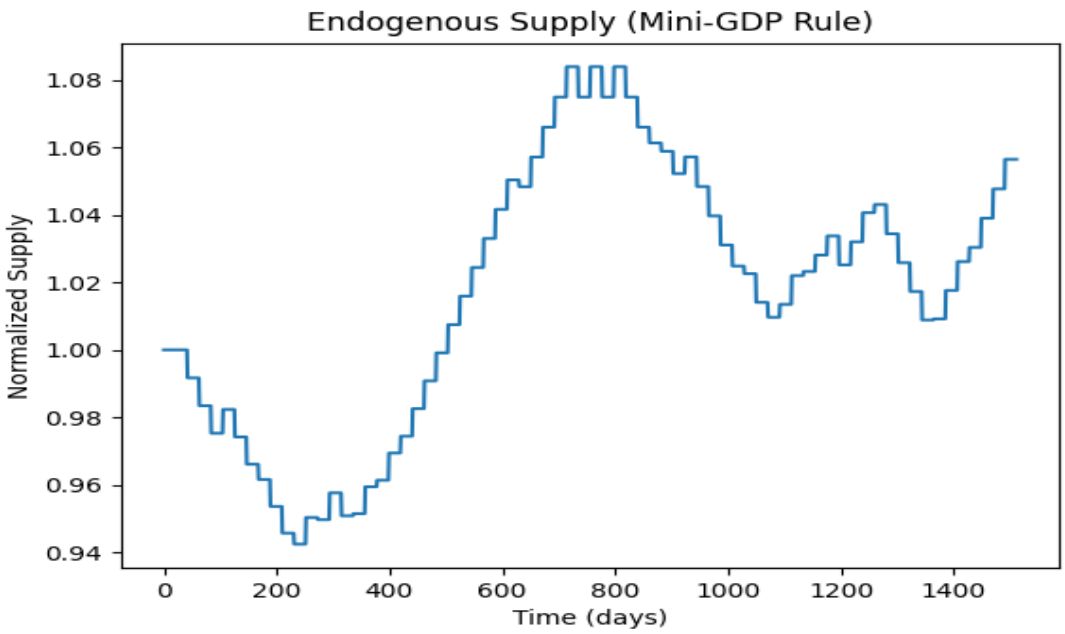
Member weights follow GDP_i^λ normalization. With $\lambda = 0.5$, concentration is reduced while preserving scale relevance. The resulting distribution prevents structural dominance and improves institutional balance.

2. Clearing-Based Mini-GDP Rule

Clearing volume V_t defines mini-GDP growth: $g_t = \log(V_t / V_{t-1})$. An EMA smoothing filter stabilizes the signal. Supply updates follow $S_{t+1} = S_t * \exp(\beta\, g_{ema})$, subject to an annual cap constraint.

Simulation Results





Conclusion

The clearing-based endogenous supply mechanism preserves structural drift control, maintains bounded supply expansion, and ensures close price-fundamental alignment. This architecture offers an operationally realistic alternative to GDP-anchored monetary frameworks.

Macroeconomic Stability of Emerging Blocs: Empirical Evidence from BRICS (2000–2024)

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Abstract

This article empirically evaluates whether aggregating real output within an economic bloc reduces growth volatility when compared to member economies taken individually. The conceptual base uses the World Bank real GDP indicator in constant 2015 USD (NY.GDP.MKTP.KD) and measures volatility as the sample standard deviation of annual real GDP growth. We compare the individual series with two bloc aggregates: (i) simple aggregation and (ii) aggregation with λ-parameterized weights (λ), fixed at a base year, to assess the trade-off between concentration and stability. The reported results indicate that the bloc's aggregated growth is more stable than any individual member and exhibits no years of negative growth in 2001–2024, while several members experience contraction years. A sensitivity analysis suggests that higher λ increases the weight of the largest economy, raising the aggregate mean growth but also increasing volatility. We conclude that multipolar aggregation confers a statistical property of smoothing idiosyncratic shocks, which is relevant for designing bloc-level macroeconomic rules anchored in aggregated output, conditional on appropriate institutional arrangements.

Keywords: BRICS; volatility; real growth; aggregation; lambda weights; macroeconomic stability.

1. Introduction

Economic bloc arrangements face a classic dilemma: gains in efficiency and scale can come with political tensions associated with asymmetric shocks and fiscal sovereignty. The Optimum Currency Areas (OCA) literature emphasizes that the feasibility of common rules depends on the capacity to absorb idiosyncratic shocks without generating unacceptable distributional costs. A central — and testable — empirical point is that simply aggregating economies can reduce the volatility of real output growth by diluting country-specific shocks. This article tests this hypothesis for the historical BRICS core (Brazil, Russia, India, China, and South Africa) over 2000–2024 using public World Bank data.

Research question: Is the bloc’s aggregated real growth more stable than the real growth of its members considered individually?

2. Data

The base series is the World Development Indicators (WDI) indicator "GDP (constant 2015 US$)" (code NY.GDP.MKTP.KD), which reports GDP levels in constant prices (base year 2015). The annual sample 2000–2024 contains 25 level observations and therefore 24 annual growth observations (2001–2024). Although BRICS has recently expanded membership, the analysis is restricted to the historical five-economy core for temporal consistency.

3. Methodology

3.1 Growth and volatility

For each country i we define annual real GDP growth as:

$g_{i,t} = (Y_{i,t} / Y_{i,t-1}) - 1$.

Volatility is the sample standard deviation of $g_{i,t}$ over 2001–2024.

3.2 Bloc aggregation

We define two bloc-level aggregates in levels:

Simple aggregate: $Y_{B,t}(0) = \Sigma_i Y_{i,t}$.

λ-weighted aggregate (weights fixed at the base year): $w_i(\lambda) = Y_{i,base}^\lambda / \Sigma_j Y_{j,base}^\lambda$, and $Y_{B,t}(\lambda) = \Sigma_i w_i(\lambda) Y_{i,t}$.

Aggregate growth is defined analogously: $g_{B,t}(\cdot) = (Y_{B,t}(\cdot)/Y_{B,t-1}(\cdot)) - 1$.

3.3 Mechanism: variance decomposition and correlations

To raise academic clarity, it is useful to state the statistical mechanism behind the 'volatility reduction'. Let aggregate growth be a linear combination: $g_{B,t} = \Sigma_i w_i g_{i,t}$. Then aggregate variance decomposes as:

$Var(\Sigma_i w_i g_i) = \Sigma_i w_i^2 Var(g_i) + 2 \Sigma_{i<j} w_i w_j Cov(g_i, g_j)$.

This expression shows why volatility can fall: when covariances are below the product of volatilities (correlations below 1) and weights are not extreme, the cross term does

not scale one-for-one with the diagonal term, allowing the aggregate to smooth idiosyncratic shocks. Therefore, a correlation matrix Corr(g_i, g_j) is a natural complement: it informs how synchronized cycles are and how much macro-risk diversification the bloc can deliver.

3.4 F-test (variance comparison)

As a simple check, we compare each country’s growth variance with the variance of the simple aggregate using a two-sided F-test (annual samples are short and should be interpreted with caution).

4. Results

4.1 Descriptive statistics (2001–2024)

Entity	Mean	Volatility	Minimum	Maximum
BRICS (simple aggregate)	6.46%	2.22 p.p.	0.31%	11.07%
BRICS (λ=0.6)	7.08%	2.36 p.p.	1.19%	12.18%
South Africa	2.14%	2.53 p.p.	-6.17%	5.60%
China	8.18%	2.77 p.p.	2.34%	14.15%
Brazil	2.30%	2.86 p.p.	-3.55%	7.53%
India	6.36%	3.11 p.p.	-5.78%	9.69%
Russia	3.14%	3.86 p.p.	-7.80%	8.50%

Core result: the simple aggregate exhibits volatility below any individual member (2.22 p.p. vs the smallest individual volatility of 2.53 p.p.) and a positive minimum growth rate over 2001–2024, while several countries show contraction years.

4.2 Annual drops: years with negative growth (2001–2024)

Brazil: 4; Russia: 4; India: 1; South Africa: 2; China: 0; BRICS aggregate (simple): 0; BRICS aggregate (λ=0.6): 0.

4.3 Stability gain (quantified)

Average member volatility: 3.02 p.p. Simple aggregate volatility: 2.22 p.p. Relative reduction: approximately 26.5%. Relative to the most volatile member (Russia), the reduction is approximately 42.4%.

4.4 F-test (country variance vs simple aggregate variance)

A highlighted case is Russia, with an approximate variance ratio of 3.01 and $p \approx 0.011$, suggesting significantly higher variance than the aggregate in the simple test. For Brazil, India, China, and South Africa, variance ratios exceed 1 but 5% significance is not reached in the simple test, partly due to the short annual sample.

5. Sensitivity to the λ parameter (weights fixed at the base year)

λ	Mean	Volatility	China weight
0 (equal weights)	6.46%	2.22 p.p.	20.0%
0.3	6.79%	2.28 p.p.	27.8%
0.6	7.08%	2.36 p.p.	36.5%

1.0	7.39%	2.45 p.p.	48.6%
2.0	7.85%	2.63 p.p.	74.7%

Reading: higher λ increases the weight of the largest economy, raising the aggregate mean growth and increasing volatility. Institutionally, λ can be interpreted as a governance lever (concentration vs sharing).

6. Discussion and implications

The empirical result is direct: the bloc smooths. Negative shocks in some members can be offset by positive performance in others, reducing aggregate volatility. This is relevant for any proposal of bloc-level macro rules (including monetary rules anchored in aggregated output), because such rules would operate on a statistically more stable variable than a standalone national GDP measure. On the other hand, aggregate stability does not eliminate classic challenges: fiscal sovereignty and distributional legitimacy (who absorbs asymmetric shocks) and the need for institutional mechanisms (stabilization funds, coordinated fiscal rules, etc.).

7. Limitations

Annual frequency: quarterly series would better capture short-run dynamics. Measurement in constant USD: using growth rates reduces scale issues, but differences in national accounts methodologies remain. Fixed weights: time-varying weights could change results; base-year fixation isolates the effect of the λ parameter.

8. Conclusion

Based on aggregation of real output for the historical BRICS core, aggregated growth is more stable than the growth of each individual member, with an estimated relative volatility reduction around 26.5% and no years of negative growth in the aggregate over 2001–2024. Sensitivity to λ highlights a trade-off between concentration and stability, reinforcing that the bloc is the natural statistical environment for common macro rules anchored in aggregate variables.

9. BRICS elastic monetary regime: stepwise fundamental and GDP-reactive policy

The simulated "BRICS elastic" architecture introduces a temporal informational asymmetry: the real economy evolves continuously, while the monetary authority observes a "reported fundamental" in discrete windows (e.g., quarterly), producing a staircase effect. In parallel, the market price evolves continuously with noise and reconverges elastically to the fundamental via a stabilization mechanism (parameters α and K). A GDP-reactive monetary rule can be written as:

$S_{t+1} = S_t (1 + \beta g_t - \gamma 1[g_t < 0])$, where β is the emission fraction proportional to growth and γ represents countercyclical defense (burn/contraction) in recession.

Appendix A. Code to reproduce the WDI tables and compute correlations/decomposition

The script below downloads the NY.GDP.MKTP.KD indicator (GDP in constant 2015 USD) via the World Bank API v2, computes annual growth, constructs the simple aggregate and the λ-weighted aggregate with base-year-fixed weights, and outputs: (i) descriptive statistics, (ii) the correlation matrix of member growth rates, and (iii) the variance decomposition of the aggregate. Note: in the environment that generated the original PDF, direct ZIP download was disabled; the code is provided for local execution.

```
Estabilidade Macroeconomica de Blocos Emergentes: Evidencia Empirica do BRICS (2000-2024)Pagina 6
Apêndice A. Codigo para reproduzir as tabelas (WDI) e calcular
correlacoes/decomposicao
O script abaixo baixa o indicador NY.GDP.MKTP.KD (PIB em US$ constantes de 2015)
via API v2 do Banco Mundial, calcula crescimento anual, constrói o agregado simples e
o agregado com pesos lambda fixos no ano-base, e produz: (i) estatisticas descritivas,
(ii) matriz de correlacao dos crescimentos, e (iii) decomposicao de variancia do
agregado. Observacao: no ambiente que gerou este PDF, o download direto em ZIP
nao foi habilitado; o codigo e fornecido para execucao local.
# ============================================================
# Reproducao (execucao local): WDI NY.GDP.MKTP.KD (2000-2024)
# ============================================================
import io, zipfile, requests
import numpy as np
import pandas as pd
from scipy import stats
COUNTRIES = ["BRA","RUS","IND","CHN","ZAF"]
INDICATOR = "NY.GDP.MKTP.KD"          # GDP (constant 2015 US$)
BASE_YEAR = 2000
START_YEAR, END_YEAR = 2000, 2024
def download_wdi_indicator_zip(indicator: str) -> pd.DataFrame:
    # World Bank API downloadformat=csv retorna um ZIP contendo um CSV principal.
    url = f"https://api.worldbank.org/v2/en/indicator/{indicator}?downloadformat=csv"
    r = requests.get(url, timeout=120)
    r.raise_for_status()
    z = zipfile.ZipFile(io.BytesIO(r.content))
    # Encontra o CSV principal (nao Metadata).
    csv_name = [n for n in z.namelist() if n.lower().endswith(".csv") and "metadata" not in n.lo
    with z.open(csv_name) as f:
        # O arquivo tem linhas iniciais de cabecalho; o header util costuma iniciar na linha 5.
        df = pd.read_csv(f, skiprows=4)
    return df
def tidy_country_panel(df_raw: pd.DataFrame, countries: list[str]) -> pd.DataFrame:
    # Mantem apenas paises desejados e transforma colunas de anos em formato longo.
    df = df_raw[df_raw["Country Code"].isin(countries)].copy()
    year_cols = [c for c in df.columns if c.isdigit()]
    out = df[["Country Code"] + year_cols].melt(
        id_vars=["Country Code"], var_name="year", value_name="Y"
    )
    out["year"] = out["year"].astype(int)
    out = out.dropna(subset=["Y"]).sort_values(["Country Code","year"])
    return out
def growth(panel: pd.DataFrame) -> pd.DataFrame:
    panel = panel.copy()
    panel["g"] = panel.groupby("Country Code")["Y"].pct_change()
    return panel.dropna(subset=["g"])
def aggregate_simple(panel_level: pd.DataFrame) -> pd.DataFrame:
    agg = panel_level.groupby("year", as_index=False)["Y"].sum()
    agg["Country Code"] = "BRICS_SUM"
    return agg[["Country Code","year","Y"]]
def weights_lambda(panel_level: pd.DataFrame, lam: float, base_year: int) -> pd.Series:
    base = panel_level[panel_level["year"] == base_year].set_index("Country Code")["Y"]
    w = (base ** lam) / (base ** lam).sum()
```

```
    return w
def aggregate_lambda(panel_level: pd.DataFrame, lam: float, base_year: int) -> pd.DataFrame:
    w = weights_lambda(panel_level, lam, base_year)
```

```
    piv = panel_level.pivot(index="year", columns="Country Code", values="Y")
    y = (piv[w.index] * w.values).sum(axis=1)
    out = y.reset_index().rename(columns={0:"Y"})
    out["Country Code"] = f"BRICS_LAM_{lam}"
    return out[["Country Code","year","Y"]]
def stats_table(panel_g: pd.DataFrame, label_map: dict[str,str]) -> pd.DataFrame:
    def desc(g):
        return pd.Series({
            "mean": g.mean(),
            "vol": g.std(ddof=1),
            "min": g.min(),
            "max": g.max(),
            "neg_years": int((g < 0).sum()),
        })
    tab = panel_g.groupby("Country Code")["g"].apply(desc).reset_index()
    tab["name"] = tab["Country Code"].map(label_map).fillna(tab["Country Code"])
    return tab[["name","mean","vol","min","max","neg_years"]].sort_values("name")
def ftest_var(x, y):
    # Teste F bicaudal: H0 Var(x)=Var(y)
    vx, vy = np.var(x, ddof=1), np.var(y, ddof=1)
    if vx >= vy:
        F = vx / vy
        dfn, dfd = len(x)-1, len(y)-1
    else:
        F = vy / vx
        dfn, dfd = len(y)-1, len(x)-1
    p = 2 * min(stats.f.cdf(F, dfn, dfd), 1 - stats.f.cdf(F, dfn, dfd))
    return F, p
# 1) Baixa e organiza dados
raw = download_wdi_indicator_zip(INDICATOR)
panel = tidy_country_panel(raw, COUNTRIES)
panel = panel[(panel["year"] >= START_YEAR) & (panel["year"] <= END_YEAR)]
# 2) Constroi agregados de nivel e calcula crescimento
agg0 = aggregate_simple(panel)
agg06 = aggregate_lambda(panel, lam=0.6, base_year=BASE_YEAR)
panel_all = pd.concat([panel, agg0, agg06], ignore_index=True).sort_values(["Country Code","year
panel_g = growth(panel_all)
panel_g = panel_g[(panel_g["year"] >= START_YEAR+1) & (panel_g["year"] <= END_YEAR)]
# 3) Matriz de correlacao dos crescimentos (membros)
piv_g = panel_g[panel_g["Country Code"].isin(COUNTRIES)].pivot(index="year", columns="Country Co
corr = piv_g.corr()
# 4) Decomposicao de variancia do agregado com pesos lambda
w = weights_lambda(panel, lam=0.6, base_year=BASE_YEAR)
cov = piv_g.cov()
var_diag = float((w**2 * np.diag(cov.loc[w.index, w.index])).sum())
var_cross = float(2 * sum(w[i]*w[j]*cov.loc[i,j] for i in w.index for j in w.index if i < j))
var_total = var_diag + var_cross
# 5) Estatisticas e teste F
labels = {
    "BRA":"Brasil","RUS":"Russia","IND":"India","CHN":"China","ZAF":"Africa do Sul",
    "BRICS_SUM":"BRICS (agregado simples)","BRICS_LAM_0.6":"BRICS (lambda=0.6)"
}
tab = stats_table(panel_g, labels)
# F-test: cada pais vs agregado simples
g_agg = panel_g[panel_g["Country Code"]=="BRICS_SUM"]["g"].values
ft_rows = []
for c in COUNTRIES:
    g_c = panel_g[panel_g["Country Code"]==c]["g"].values
```

```
    F, p = ftest_var(g_c, g_agg)
    ft_rows.append((labels[c], F, p))
ft = pd.DataFrame(ft_rows, columns=["pais","F","p_value"])
print(tab)
print(corr)
print({"var_diag":var_diag, "var_cross":var_cross, "var_total":var_total})
print(ft)
```

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